OSPF States:

The OSPF (Open Shortest Path First) routing protocol has a well-defined process for establishing neighbor relationships and synchronizing link-state information between routers.

1. Down State

2. Init State

3. Two-Way State

4. ExStart State

5. Exchange State

6. Loading State

7. Full State

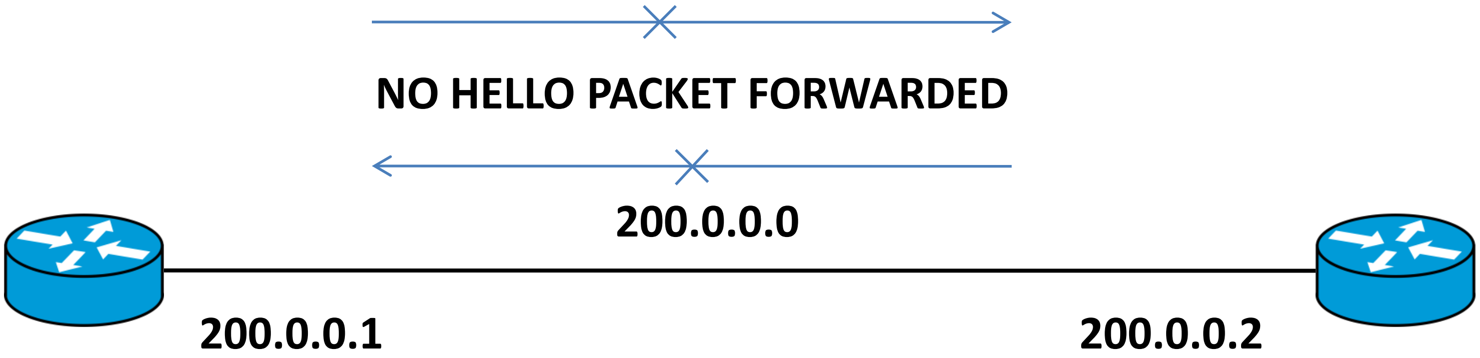
Tables in OSPF:

Neighbor Table: Tracks OSPF neighbors and their state (formed during the Init and Two-Way states). It Contains list of all OSPF neighbors.

Link-State Database (LSDB): Contains the entire network topology and is used to calculate the best paths (formed and updated with LSAs).

Routing Table: Lists the best routes to each destination network based on the information in the LSDB (formed after SPF calculations).

OSPF States: Down State

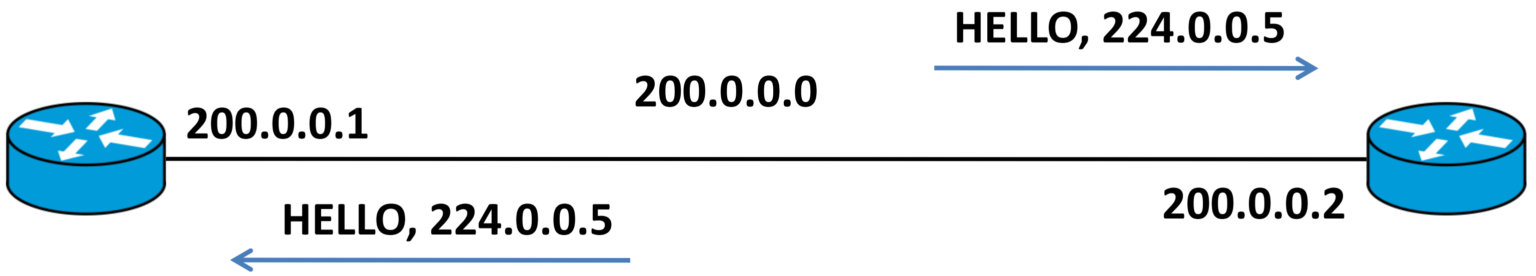


Down State:

Description: The initial state of the OSPF process. In this state, a router has not received any OSPF Hello packets from a neighbor on an interface.

Transition: The router transitions out of the Down state when it receives a Hello packet from another OSPF router

OSPF State: Init State

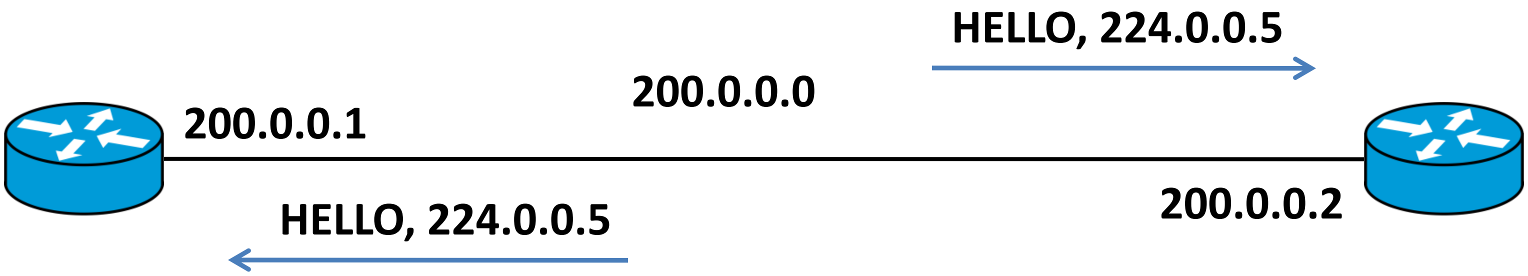


Init State:

Description: In this state, the router has received a Hello packet from a neighbor, but the Hello packet does not contain the router's own Router ID in the neighbor list. This indicates that the neighbor is not aware of this router yet. Neighbor table will be created.

Transition: The router will move to the next state when it receives a Hello packet that includes its own Router ID in the neighbor list, indicating that the neighbor recognizes it.

OSPF State: Init State



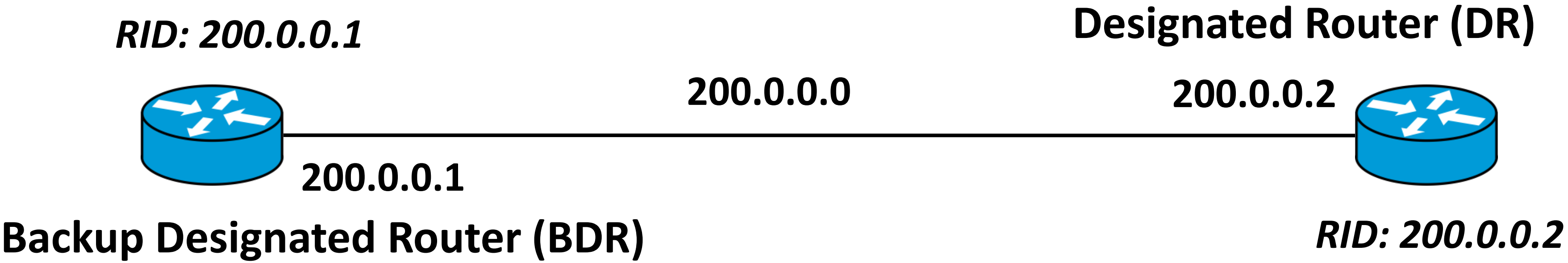
Capturing from - [R1 FastEthernet0/0 to R2 FastEthernet0/0]

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	200.0.0.1	224.0.0.5	OSPF	90	Hello Packet
2	2.021105	ca:02:15:78:00:08	ca:02:15:78:00:08	LOOP	60	Reply
3	2.037062	ca:01:11:b0:00:08	ca:01:11:b0:00:08	LOOP	60	Reply
4	5.676333	ca:02:15:78:00:08	CDP/VTP/DTP/PAgP/UDLD	CDP	366	Device ID: R2
5	9.599842	200.0.0.1	224.0.0.5	OSPF	90	Hello Packet

OSPF State: Two-Way State

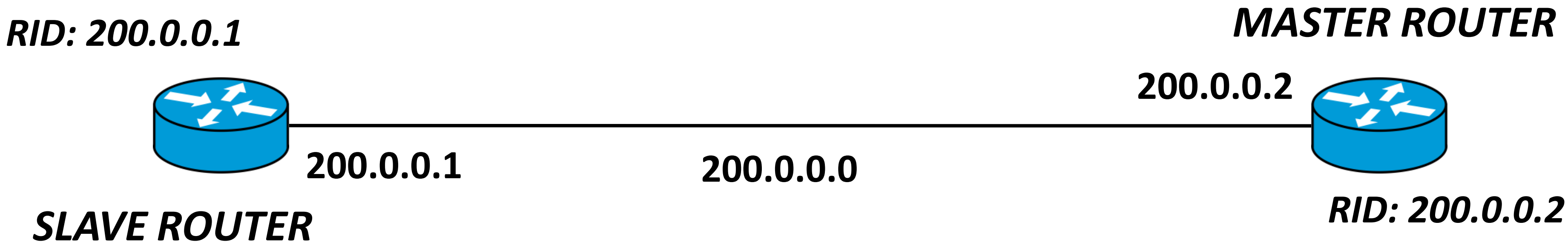


Two Way State:

Description: This state indicates that bidirectional communication is established between two routers. The routers have seen each other's Router IDs in the Hello packets they exchanged.

Transition: At this stage, if the network is a broadcast or non-broadcast multi-access (NBMA) network, the routers will elect a Designated Router (DR) and a Backup Designated Router (BDR).

OSPF State: ExStart State

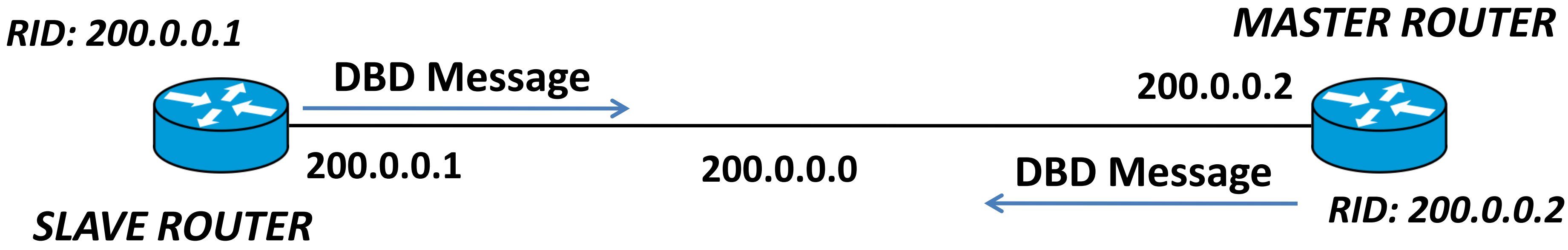


Exstart State:

Description: The routers begin to establish a master-slave relationship to synchronize their link-state databases. The master router controls the exchange of DBD (Database Description) packets, and the slave router follows.

Transition: The router with the higher Router ID becomes the master, and the other becomes the slave. Once the master-slave relationship is determined, the routers transition to the next state.

OSPF State: Exchange State

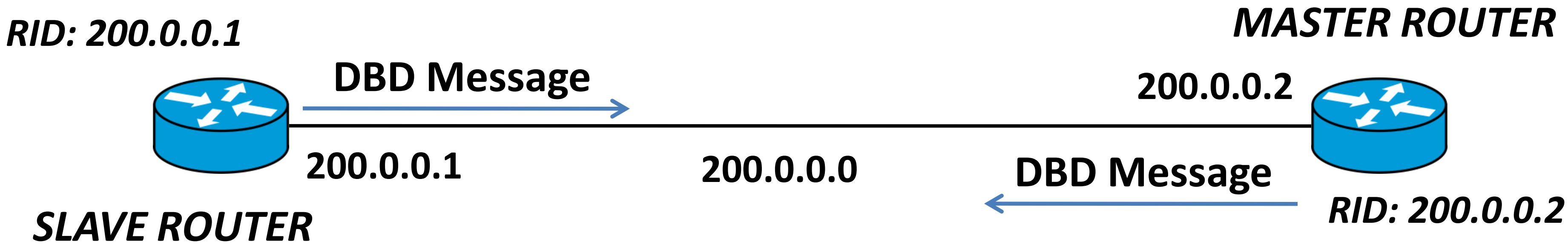


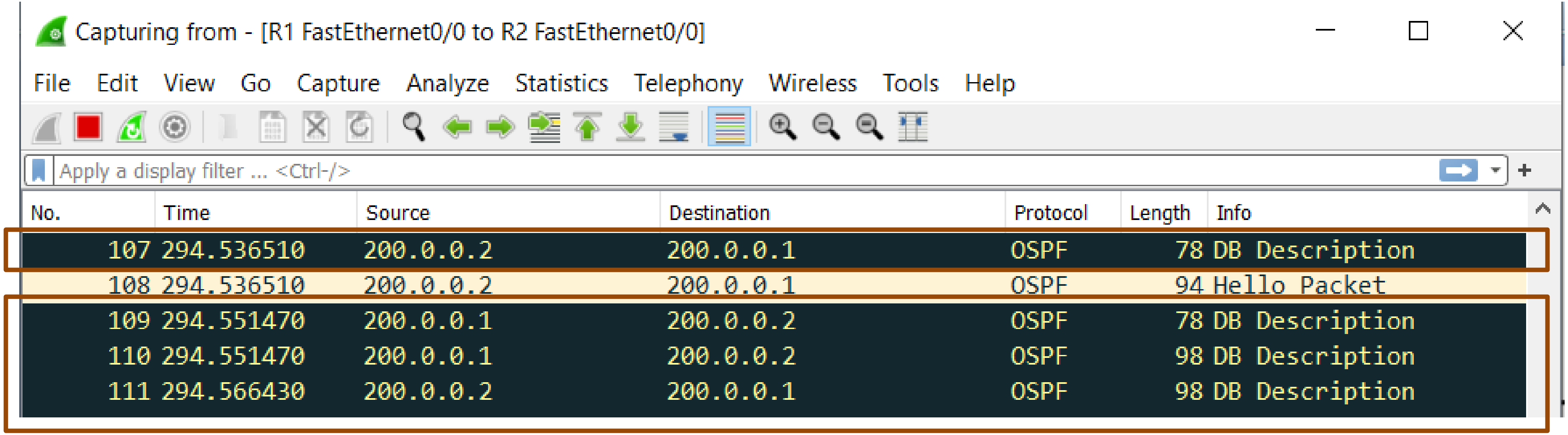
Exchange State:

Description: The routers exchange DBD packets containing summaries of the LSAs in their respective link-state databases. This allows the routers to identify which LSAs they need from each other.

Transition: If the routers have more LSAs to exchange, they continue to send DBD packets. When both routers have sent and received all necessary DBD packets, they move to the next state.

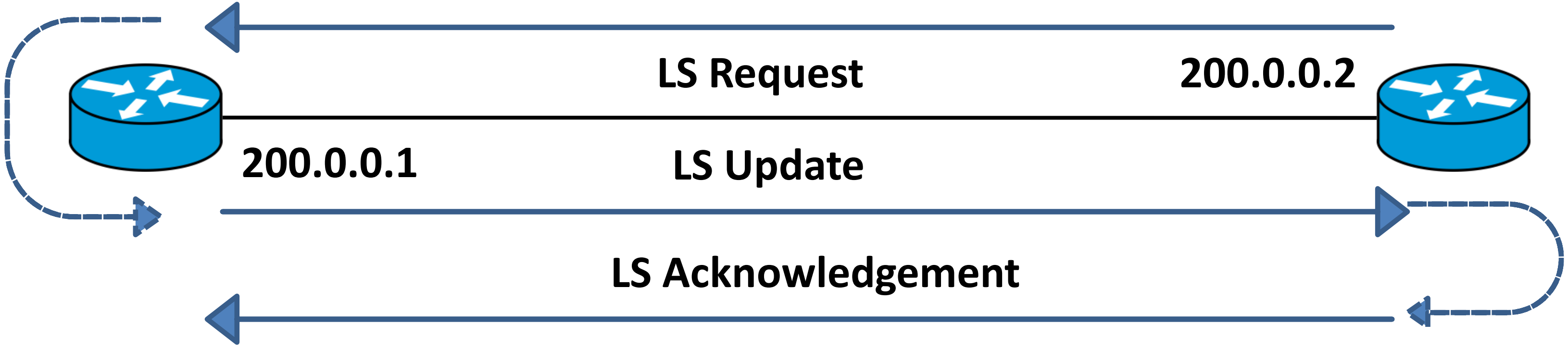
OSPF State: Exchange State





No.	Time	Source	Destination	Protocol	Length	Info
107	294.536510	200.0.0.2	200.0.0.1	OSPF	78	DB Description
108	294.536510	200.0.0.2	200.0.0.1	OSPF	94	Hello Packet
109	294.551470	200.0.0.1	200.0.0.2	OSPF	78	DB Description
110	294.551470	200.0.0.1	200.0.0.2	OSPF	98	DB Description
111	294.566430	200.0.0.2	200.0.0.1	OSPF	98	DB Description

OSPF State: Loading State

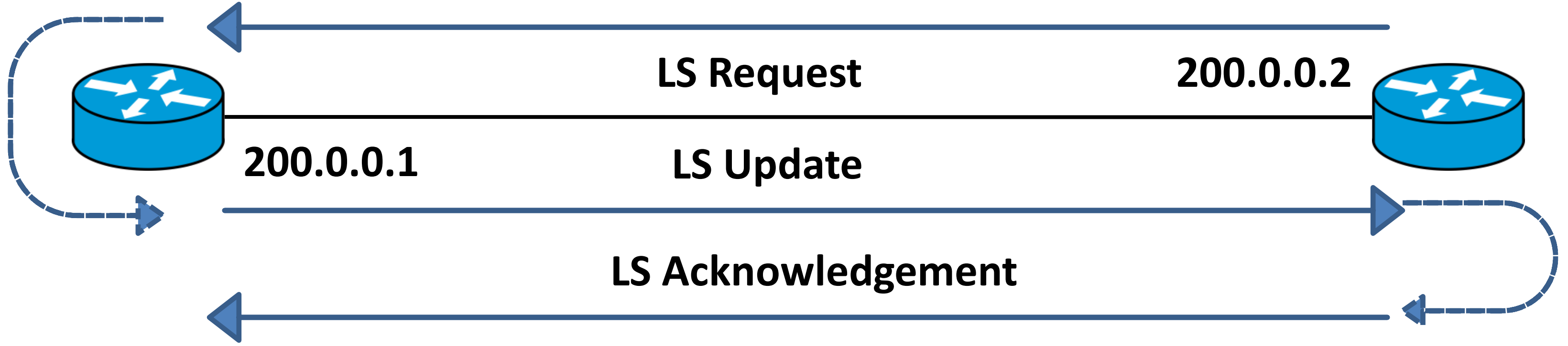


Loading State:

Description: The routers exchange DBD packets containing summaries of the LSAs in their respective link-state databases. This allows the routers to identify which LSAs they need from each other.

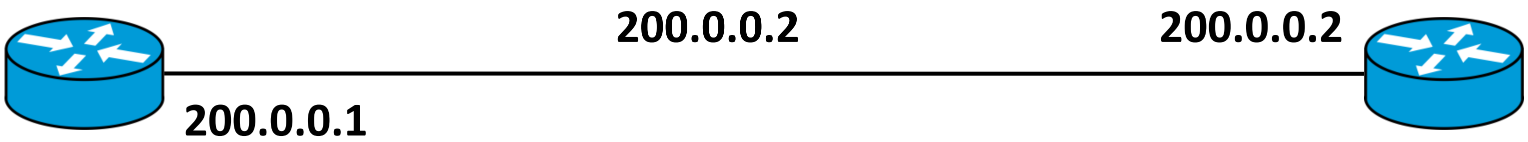
Transition: If the routers have more LSAs to exchange, they continue to send DBD packets. When both routers have sent and received all necessary DBD packets, they move to the next state.

OSPF State: Loading State



112	294.582387	200.0.0.1	200.0.0.2	OSPF	70 LS Request
113	294.582387	200.0.0.1	200.0.0.2	OSPF	78 DB Description
114	294.598344	200.0.0.2	200.0.0.1	OSPF	98 LS Update
115	294.598344	200.0.0.2	200.0.0.1	OSPF	70 LS Request
116	294.614302	200.0.0.1	200.0.0.2	OSPF	98 LS Update
117	295.031187	200.0.0.2	224.0.0.5	OSPF	98 LS Update
118	295.092024	200.0.0.1	224.0.0.5	OSPF	98 LS Update
119	295.122942	200.0.0.1	224.0.0.5	OSPF	94 LS Update
120	296.405513	200.0.0.2	224.0.0.5	OSPF	94 Hello Packet
121	297.081705	ca:01:11:b0:00:08	CDP/VTP/DTP/PAgP/UDLD	CDP	366 Device ID: R1 Port
122	297.113621	200.0.0.1	224.0.0.5	OSPF	98 LS Acknowledge
123	297.129578	200.0.0.2	224.0.0.5	OSPF	118 LS Acknowledge

OSPF State: Full State



Routing Table will be updated

Full State:

Description: This is the final state where the routers have fully synchronized their link-state databases. The routers have exchanged all necessary information, and the adjacency is fully established.

Result: In this state, the routers can participate in OSPF routing by using the synchronized information to calculate the best paths through the network.